SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** 26177
- **Problem Statement Title –** A deployable AI-powered autonomous drone that aids search-and-rescue operations by detecting people and hazards, thereby improving responder safety and reducing victim discovery time
- **Theme –** Disaster Management
- **PS Category –** Hardware
- **Team ID –** SIH26-AOH-T326
- **Team Name (Registered on portal) –** 200 OK

200 OK

ResQ-Edge: Offline Edge-AI Autonomous Drone for Disaster Response

SMART INDIA HACKATHON 2026

Current Challenges:

1. **Delayed Discovery** – Survivors take time to locate in vast disaster zones.
2. **Rescuer Safety Risks** – Collapsed structures, fire, floods, and debris endanger teams.
3. **Visibility Limitations** – Darkness, smoke, and obstructions conceal survivors.
4. **Communication Breakdowns** – Network-dependent systems fail during outages.
5. **Limited Situational Awareness** – Raw drone footage lacks actionable insights.

Our Idea – ResQ-Edge:

An **autonomous edge-AI drone** that detects survivors and hazards, maps disaster zones, and delivers prioritized rescue intelligence even without internet connectivity.

Proposed Solution – ResQ-Edge:

- Edge AI Processing
- Sensor Fusion (RGB + Thermal)
- Survivor Detection
- Hazard Detection
- Geo-Mapping

- Risk Prioritization
- Autonomous Navigation
- Offline Communication
- Command Dashboard

Key Differentiator:

Aspect	Traditional Drone	ResQ-Edge
Raw video	Raw video	Actionable intelligence
Cloud-dependency	Cloud-dependent	Edge AI processing
Data interpretation	Manual interpretation	Automated detection and prioritization
Situational awareness	Limited situational awareness	Offline-capable operation

Core Flow:

Drone Sensors → Edge AI → Survivor/Hazard Detection → Risk Assessment → Geo-tagged Map → Rescue Team

200 OK

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

SENSOR LAYER
- RGB Camera
- Thermal Camera
- GPS/GNSS
- IMU
- LiDAR / Depth Sensor

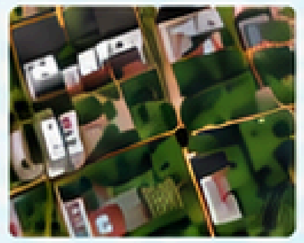
EDGE AI PROCESSING
- Person Detection
- Hazard Recognition
- RGB-Thermal Fusion
- Real-time Classification

LOCALIZATION AND MAPPING
- GPS / SLAM
- Geo-tagging
- Occupancy Map
- Drone Trajectory

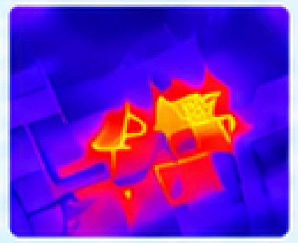
DECISION AND MISSION ENGINE
- Risk Scoring
- Priority Assignment
- Route Planning
- Mission Management

COMMAND CENTER
- Disaster Map
- Survivor Alerts
- Hazard Alerts
- Mission Status
- Incident Reports

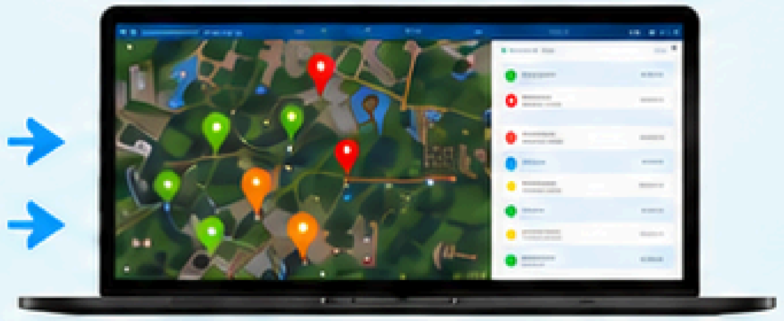
AUTONOMOUS SEARCH



RGB VIEW



THERMAL VIEW



REAL-TIME DISASTER INTELLIGENCE

AI PROCESSING PIPELINE
1. Camera Frame
2. Pre-processing
3. AI Model Inference
4. Bounding Boxes / Segmentation
5. Confidence Filtering
6. RGB-Thermal Fusion
7. GPS Association
8. Hazard Correlation
9. Risk Scoring
10. Priority Assignment
11. Alert Generation

HARDWARE STACK
- Drone
- Flight Controller
- Edge AI Computer
- RGB Camera
- Thermal Camera
- GPS/ GNSS
- IMU
- LiDAR / Depth Sensor
- Power Management System
- Communication Module

SOFTWARE STACK
- PX4 / ArduPilot
- ROS 2
- YOLO
- ONNX Runtime / TensorRT
- OpenCV
- MAVLink / MAVSDK
- Node.js / Express
- React
- Leaflet / Mapbox
- SQLite / PostgreSQL

200 OK

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Innovation

Offline edge AI
- On-drone inference and local storage
- Alerts without cloud dependency

RGB + thermal fusion
- Visual and heat-signature detection
- Reliable operation in darkness and low visibility

Risk-aware prioritization
- Contextual survivor and hazard alerts
- Scoring by presence, severity, accessibility

Technical feasibility
- Off-the-shelf drones and sensors
- Proven AI and open-source robotics
- Edge-optimized inference
- Modular hardware and software design

Prototype feasibility
- Mock disaster demo with autonomous grid search
- RGB and thermal survivor detection
- Hazard detection: fire, flood, debris
- GPS geo-tagging and offline dashboard
- Return-to-home safety

Aspect	Details
Cost	Uses commercially available drone and edge AI hardware
Scalability	Architecture supports multi-drone deployment
Real-world Impact	Significantly reduces search time and saves lives
Adoption Potential	Applicable in disaster response, defense, and industrial safety
Sustainability	Works in low or no connectivity environments
Alignment with SIH	Supports Atmanirbhar Bharat, indigenous innovation and national disaster preparedness goals

200 OK

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

Impacts

- **Accelerated survivor discovery –** Autonomous aerial scanning reduces search time.
- **Enhanced responder safety –** Hazards assessed remotely before entry.
- **Actionable rescue decisions –** Survivor and hazard data with confidence and risk priorities.
- **Offline continuity –** AI detection and logging without internet.
- **Future expansion –** Multi-drone coordination, GPS-denied navigation, advanced sensors.

Benefits

- **Situational awareness –** Disaster maps, alerts, mission status improve planning.
- **Multi-disaster coverage –** Floods, cyclones, quakes, landslides, collapses, fires, industrial emergencies, urban SAR.
- **Broad user base –** Agencies, fire/rescue, SAR teams, emergency orgs, industrial safety, humanitarian groups.
- **Scalable prototype –** From single drone demo to AI-generated reports and automated rescue routes.
- **Core USP –** Converts sensor data into actionable rescue intelligence, not just footage.

Target users

- Disaster Mgmt Agencies
- Fire and Rescue Departments
- Search and Rescue Teams
- Emergency Response Organizations
- Industrial Emergency Teams
- Humanitarian Organizations

Core USP | *"ResQ-Edge does not just capture disaster footage. It transforms sensor data into actionable rescue intelligence."*

200 OK

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

1. **FPGA-based UAV and UGV for search and rescue applications: A case study**
   - **Source:** ScienceDirect, 2024
   - **Link:** https://www.sciencedirect.com/science/article/abs/pii/S004579062400418X

2. **UAV-Based Visual Detection and Tracking of Drowning Victims in Maritime Rescue Operations**
   - **Source:** Drones (MDPI), 2026
   - **Link:** https://doi.org/10.3390/drones10020146

3. **Seeing Through Sparse Foliage: Quality–Occlusion-Guided RGB–Thermal Fusion for Drone-Based Person Detection**
   - **Source:** Remote Sensing (MDPI), 2026
   - **Link:** https://doi.org/10.3390/rs18050774

4. **Edge-Driven Real-Time UAV Thermal Imaging for Survivor Identification**
   - **Source:** Springer Professional, 2026
   - **Link:** https://www.springerprofessional.de/en/edge-driven-real-time-uav-thermal-imaging-for-survivor-identific/51527334

5. **Real-time Human Detection in Fire Scenarios using Infrared and Thermal Imaging Fusion**
   - **Source:** arXiv, 2023
   - **Link:** https://arxiv.org/pdf/2307.04223

6. **Deep Learning with RGB and Thermal Images Onboard a Drone for Monitoring Operations**
   - **Source:** Journal of Field Robotics (Wiley), 2022
   - **Link:** https://onlinelibrary.wiley.com/doi/full/10.1002/rob.22082

7. **COXNet: Cross-Layer Fusion with Adaptive Alignment and Scale Integration for RGBT Tiny Object Detection**
   - **Source:** arXiv, 2025
   - **Link:** https://arxiv.org/pdf/2508.09533

8. **Autonomous UAVs as Rescue Agents: Blink Detection for Human-State-Aware Survivor Localization**
   - **Source:** Drones (MDPI), 2026
   - **Link:** https://www.mdpi.com/2504-446X/10/6/417